03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 1

(JCD 1.1)

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UNIVERSITY OF BIRMINGHAM

Outline

Course structure

Relevance of stellar structure and evolution

Stellar Timescales
- Dynamical timescale
- Thermal timescale
- Nuclear timescale

Summary

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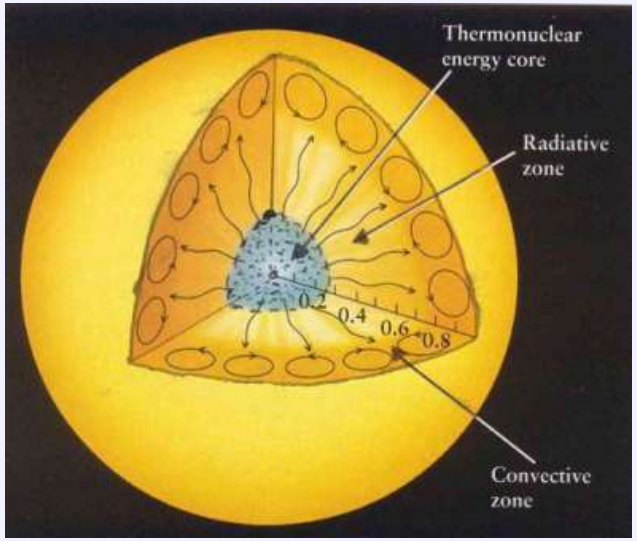
- Lectures: Mon 11am & Wed 12pm every week.
- Slides will be available on Canvas before the lecture
  - See Canvas for script to make your own handouts
- Recording and visualiser scan available on Canvas after the lecture
- Assessment:
  - Two sets of assessed problems (10% each)
  - One written 1.5hr exam (80%) - no question choice
- Two supervision sessions to discuss solutions to assessed problems: Tue 11 Nov and Tue 9 Dec, 11am
- Recommended reading material (available on Canvas and through library - no need for purchase):
  - Main: Lecture notes by Joergen Christensen-Dalsgaard
  - Lecture notes by Onno Pols
  - See recommended books in Resource Lists

Contacting me

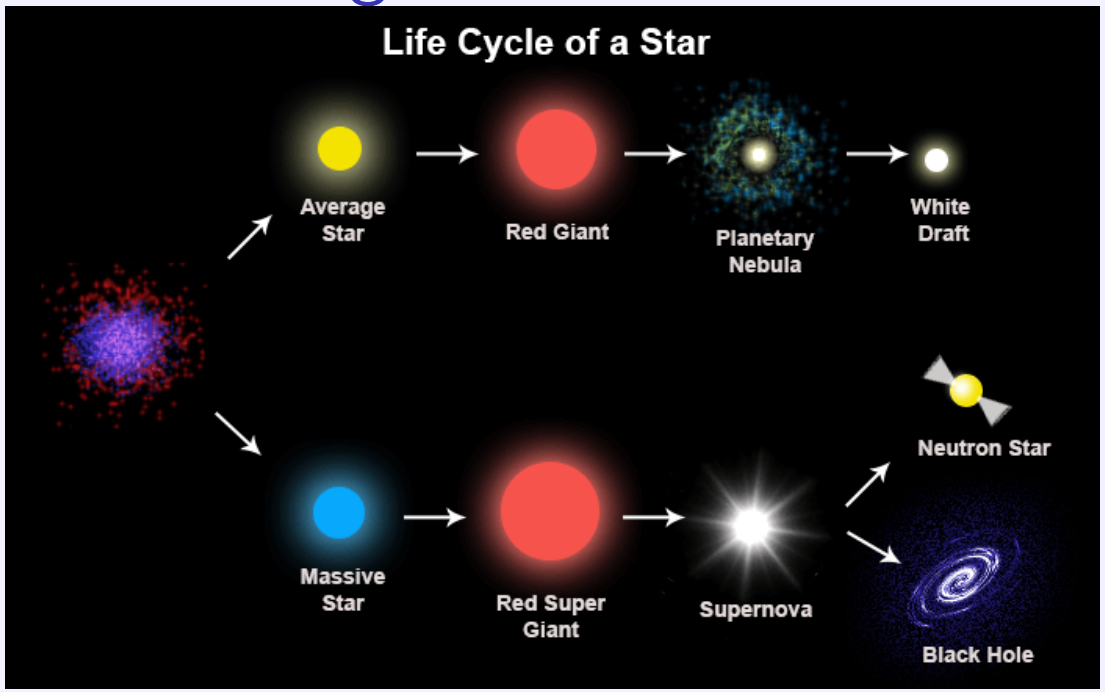
- No formal office hours.
- Contactable through:
  - Email - a.mortier@bham.ac.uk
  - Teams Chat
- If face-to-face meeting required, please book an appointment first via mechanisms above.

Part 1 - Equations of stellar structure and evolution



- Relevance, HR diagram, stellar observable properties
- Hydrostatic equilibrium, mass continuity
- Equations of state
- Virial theorem and simple models
- Energy transport: radiation
- Energy transport: convection
- Energy generation and nuclear reactions
- Full set of equations, boundary conditions and solutions

Part 2 - Qualitative stages of stellar evolution



- Star formation, approach to main sequence
- Main sequence
- Evolution beyond MS: 1 solar mass star
- Evolution beyond MS: 5 solar mass star
- Massive star evolution, supernovae
- Isochrones and stellar clusters
- End states of stars: white dwarfs, neutron stars, black holes

Slido

Join SLIDO - #4249841

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Importance of stars

It is the "astro"
in astronomy / astrophysics

History of our Galaxy / the Universe

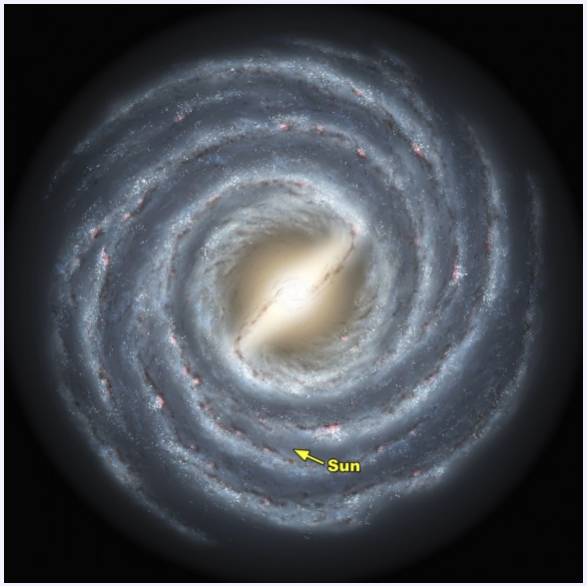
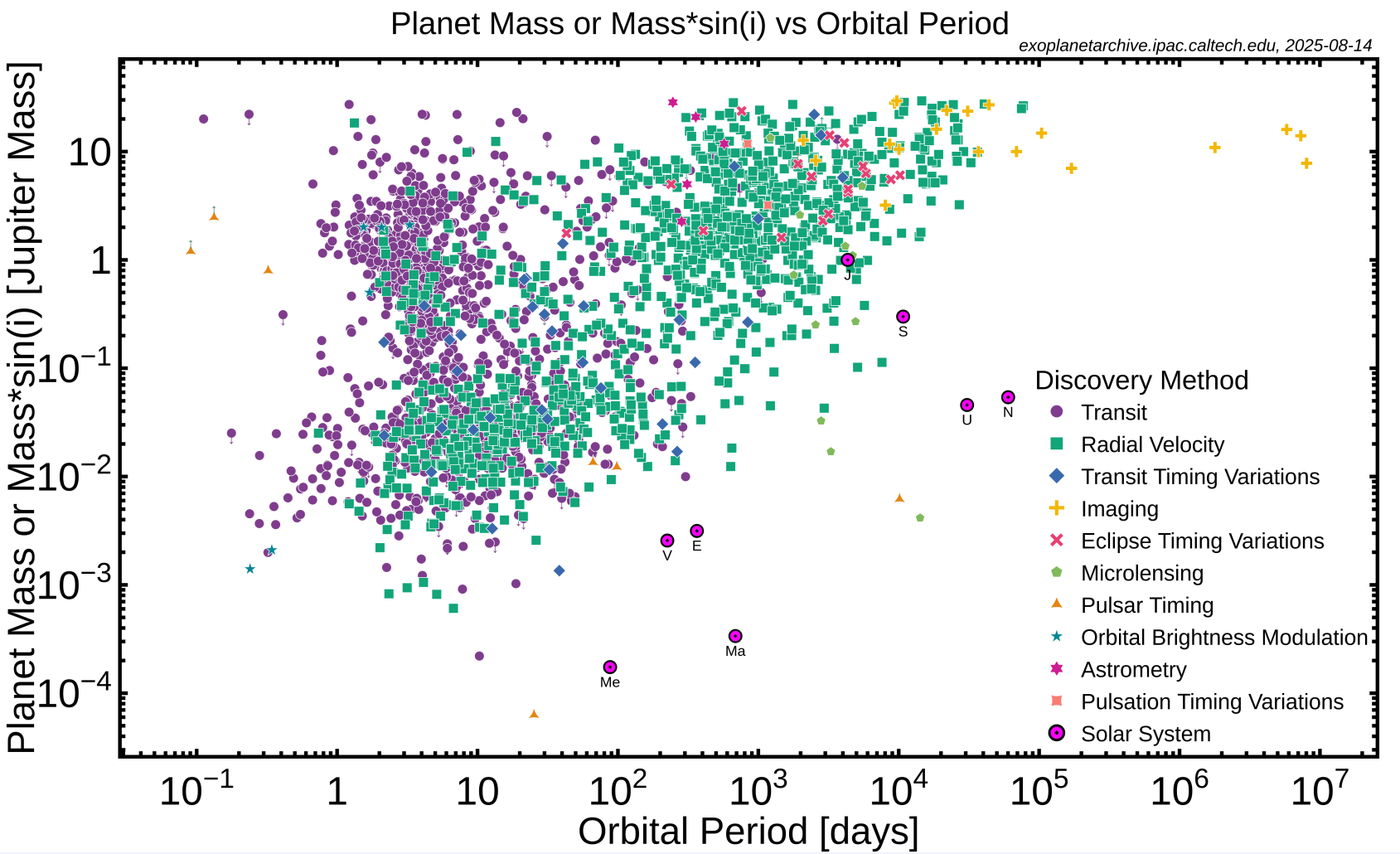


Image credit: Caltech, NASA, ESA

Exoplanets


Planet Mass or Mass*sin(i) vs Orbital Period
exoplanetarchive.ipac.caltech.edu, 2025-08-14
Planet Mass or Mass*sin(i) [Jupiter Mass]
Orbital Period [days]
Discovery Method
Transit
Radial Velocity
Transit Timing Variations
Imaging
Eclipse Timing Variations
Microlensing
Pulsar Timing
Orbital Brightness Modulation
Astrometry
Pulsation Timing Variations
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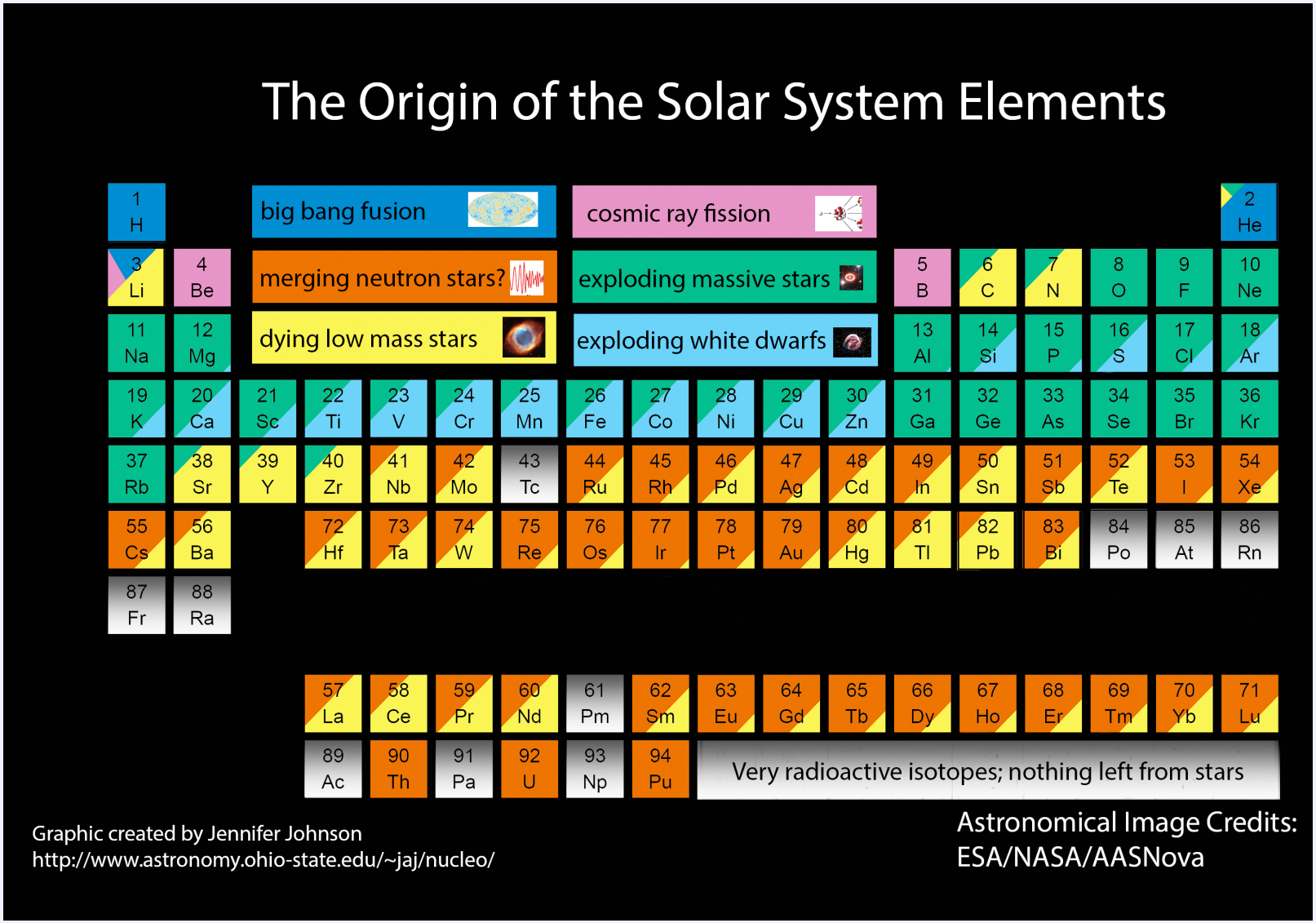
Origin of all elements and thus life



Stars are the labs of stellar physics

Image credit: University of Birmingham - GW group

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Dynamical timescale

- Time it would take for a star to collapse completely if there were no pressure forces (or explode if there were no gravity).
- Simple estimate: time it takes for a particle to fall freely from the surface to the centre of a star.
- Remember Mechanics!

$$R = \frac{1}{2} g \, t_{\mathrm{dyn}}^2$$

$$\Downarrow$$

$$t_{\mathrm{dyn}} = \sqrt{\frac{2R}{g}} = \sqrt{\frac{2R^3}{GM}}$$

Dynamical timescale of the Sun

$$t_{\mathrm{dyn},\odot} = \sqrt{\frac{2R_\odot^3}{GM_\odot}} \approx 2000\,\mathrm{s} \approx 0.5\,\mathrm{hr}$$

- Dynamical timescales very short, many magnitudes shorter than typical stellar ages.
- Consequence: Assumption of hydrostatic equilibrium for MS stars is adequate - the forces are very nearly balanced.
- Alternative definition: Timescale over which a star responds to deviations from hydrostatic equilibrium.
- Use scaling relation:

$$t_{\mathrm{dyn},*} = 0.5\,\mathrm{hr}\left(\frac{R_*}{R_\odot}\right)^{3/2}\left(\frac{M_*}{M_\odot}\right)^{-1/2}$$

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Thermal time scale = Kelvin-Helmholtz timescale

- Definition: τ_{th}
  - The time over which the thermal energy of a star can supply its luminosity.
  - The timescale for the release of gravitational energy.
- Gravitational binding energy of a star:

$$\Omega = -\frac{GM^2}{R}$$

- Luminosity L: the amount of energy it radiates per unit time:

$$L \sim \frac{-\Omega}{\tau_{th}}$$

$$\Downarrow$$

$$\tau_{th} \sim \frac{-\Omega}{L} = \frac{GM_*^2}{R_* L}$$

Thermal timescale for the Sun

- Example: For the Sun: $\tau_{th} \sim 3 \cdot 10^7\,\mathrm{yr}$
- Use scaling relation:

$$\tau_{th,*} = 30\,\mathrm{Myr} \left(\frac{M_*}{M_\odot}\right)^2 \left(\frac{R_*}{R_\odot}\right)^{-1} \left(\frac{L_*}{L_\odot}\right)^{-1}$$

- But the Sun (and the Earth) has an age $> 10^9\,\mathrm{yr}$
- Slow gravitational contraction alone is not sufficient as energy source (though it is important at some stages of stellar evolution)
- A much more powerful energy source is needed $\rightarrow$ nuclear energy?

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Nuclear timescale

- Most energy comes from hydrogen fusion: 4 protons become 1 helium nucleus
- Rough energy release calculation:

$$4\,{}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + \text{energy}$$

$$4\,(1.008\,\text{amu}) \rightarrow (4.004\,\text{amu}) + \text{energy}$$

- Thus 0.028 amu is converted into energy: i.e. 0.7% of each hydrogen is converted to energy:

$$\Delta E = \Delta m\, c^2 \approx 7 \cdot 10^{-3}\, M_*\, c^2$$

- Nuclear reaction only occurs in $\sim 10\%$ inner mass of star.

$$\Downarrow$$

$$t_{\text{nuc}} = 7 \cdot 10^{-4}\, \frac{M_*\, c^2}{L_*}$$

Nuclear timescale for the Sun

- For the Sun: $t_{\rm nuc} \approx 10^{10}\,{\rm yr}$
- Use scaling relation:

$$t_{\rm nuc,*} = 10\,{\rm Gyr}\left(\frac{M_*}{M_\odot}\right)\left(\frac{L_*}{L_\odot}\right)^{-1}$$

- H burning is largest part of star's life. Nuclear timescale good estimate of stellar age.
- Stellar age a steeply decreasing function of mass

Summary

- Knowledge of stellar physics crucial across all of astronomy
- Dynamical: $t_{\mathrm{dyn}} = \sqrt{\frac{2R^3}{GM}}$
- Thermal: $\tau_{th} = \frac{GM_*^2}{R_* L}$
- Nuclear: $t_{\mathrm{nuc}} = 7 \cdot 10^{-4} \frac{M_* c^2}{L_*}$
- $t_{\mathrm{dyn}} < \tau_{th} < t_{\mathrm{nuc}}$

NEXT LECTURE: HR diagram, the life of a star